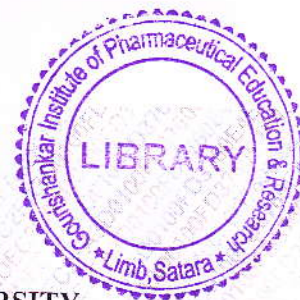




DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY

LONERE – RAIGAD – 402 103

Winter Semester Examination – December – 2019

Course: B.Pharm.

Semester: IV

Subject with Subject Code: Physical Pharmaceutics-II (BP403T)

Marks: 75

Date: 09/12/2019

Duration: 3hrs

Time: 10.30am to 1.30 pm

Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q.No.1 Attempt the following questions (All Questions Compulsory) (20 Marks)

- 1) Flocculated suspensions exhibit the flow of a type.....
 - A. Dilatants
 - B. Newtonian
 - C. Plastic
 - D. Pseudoplastic
- 2) Andreasen apparatus consists of.....
 - A. Balance
 - B. Electrode
 - C. Hydrometer
 - D. Pipette
- 3) The size of colloidal dispersed phase is
 - A. less than 1.0 nm
 - B. between 0.5 μm to 1.0 nm
 - C. greater than 0.5 μm
 - D. none of above
- 4) $K = x/at(a-x)$ is the rate constant associated with Order reaction.
 - A. Zero
 - B. First
 - C. Second
 - D. Pseudo first
- 5) A wetting agent is included in the formulation of a suspension, particularly when the suspended particles-----
 - A. Are hydrophobic
 - B. Are more denser than the vehicle
 - C. Are water soluble
 - D. Have lesser interfacial tension

- 6) is the irreversible type of colloids.
- A. Lyophilic colloids
 - B. Lyophobic colloids
 - C. Association colloids
 - D. Hydrophilic colloids
- 7) Which one of the following physical property is NOT a rheological property?
- A. Body and slip
 - B. Spreadability
 - C. Surface tension
 - D. Viscosity
- 8) When the angle of repose is between $25-30^\circ$ it will indicate that the powder material is having Flow.
- A. Excellent
 - B. Good
 - C. Passable
 - D. Poor
- 9) Porosity of a porous powder is defined as:
- A. void volume/Bulk volume
 - B. Void volume/bulk volume
 - C. Void volume/true volume
 - D. True volume/bulk volume
- 10) The degradation of solid in presence of moisture follows -----order kinetics.
- A. Zero
 - B. First
 - C. Second
 - D. Pseudo first
- 11) An 'emulsion within emulsion' is defined as-----
- A. o/w/w
 - B. w/o/o
 - C. w/o/o/w
 - D. w/o/w
- 12) require a specialized method for the preparation.
- A. Lyophilic colloids
 - B. Lyophobic colloids
 - C. Association colloids
 - D. Micelle

13) Brook-field viscometer is an example of type:

- A. Cone and plate
- B. Extrusion
- C. Rotating sphere
- D. Rotating spindle

14) In Coulter-counter, as the particles travel through the orifice, the event that occurs is.....

- A. Conductance between the electrodes increases
- B. Electronic scanners produce photographs for volume measurement
- C. Resistance between the electrodes increases
- D. Sedimentation increases

15) The HLB range of an emulsifier used in the preparation of W/O emulsion is--

- A. 3 to 6
- B. 7 to 12
- C. 13 to 15
- D. more than 15

16) The half life of substance in first order kinetics expressed by-----

- A. $0.301/K_1$
- B. $0.693/K_1$
- C. $0.105/K_1$
- D. $K_1/0.693$

17) The type of viscosity specified in I.P.(Ostwald viscometer) is:

- A. Absolute viscosity
- B. Dynamic viscosity
- C. Kinematic viscosity
- D. Viscosity coefficient

18) For an ideal suspension, the sedimentation volume should be-----

- A. Equal to one
- B. less than one
- C. Equal to one or more than one
- D. zero

19) In case of suspensions, all statements below are true EXCEPT that-----

- A. It contains a suspending agent
- B. It does not include a preservative
- C. It is an injectable preparation
- D. It is an oral preparation

20) -----shows fastest rate of degradation among dosage form.

- A. Solid
- B. Emulsion
- C. Suspension
- D. Solution

Q.No.2 Attempt any TWO questions of the following: (20 Marks)

- A) Explain Newtonian & Non Newtonian types of flow with its rheograms, mechanisms & with suitable examples.
- B) What are colloidal dispersion. Explain its types & Discuss electrical, kinetic & optical properties of colloids.
- C) Define emulsion, classify it with examples & Discuss in detail instabilities of emulsions & how is it prevented.

Q.No.3 Attempt any SEVEN questions of the following: (35 Marks)

- A) Define Micromeritics with its applications & discuss methods for determining particle size.
- B) Explain the influence of physical & chemical factors on degradation of Pharmaceuticals.
- C) Write the principle & working of Ostwald's viscometer & write applications of study of rheology.
- D) What is meant by sedimentation parameters? How are they evaluated?
- E) Define angle of repose? How it is determined? Give its importance in preparation of dosage forms.
- F) Write significance of accelerated stability testing & Discuss procedure for accelerated stability testing with any one example.
- G) Explain in detail about Thixotropy & Negative thixotropy with its curves.
- H) What is protective action of colloids. How it is determined & Enumerate the pharmaceutical applications of colloids.
- I) Define porosity with its applications in pharmacy & write note on different properties of powders.

*****END OF PAPER*****